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Trajectory-based far infrared collision-induced absorption spectra of CH_4–CO_2 in the temperature range from 70 K to 400 K☆

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ABSTRACT

This paper presents the far-infrared collision-induced absorption (CIA) spectra for CH_4–CO_2 pairs obtained using trajectory-based simulations across a broad temperature range of 70–400 K. Our calculations rely on explicit quantum-chemical calculation of the potential energy and induced dipole surfaces, constructed under the rigid-monomer approximation. Our developed trajectory-based method enables modeling of the true dimer contribution to CIA profile, which is particularly important at low temperatures. Furthermore, we introduce a significant improvement to the Monte Carlo computational scheme, which substantially accelerates its convergence in the frame of our method. The spectra obtained in this work thus provide a reliable reference for evaluating the impact of CIA in radiative transfer models, with direct application for CO_2–rich atmospheres such as Martian paleoatmosphere.

1. Introduction

The present-day Martian surface is arid and dry, but diverse geological evidence indicates that conditions on Noachian-era Mars were warmer and wetter [1]. Explaining the presence of liquid water is particularly problematic, given the lower solar luminosity at the time, making the warm climate of early Mars a major enigma of planetary science [2]. A greenhouse effect driven by reducing atmospheric conditions has been proposed, initially by Sagan [3], and in a plethora of subsequent studies. Recent climate simulations suggested that methane and hydrogen in the paleoatmosphere could have sufficiently enhanced greenhouse warming potential through CH_4–CO_2 and CO_2–H_2 collision-induced absorption (CIA) in the far infrared [4,5]. This mechanism is critical in the 200–600 cm^{-1} spectral range that forms a transparency window of CO_2-dominated atmosphere.

Initially, due to the lack of available data for CIA coefficients of CH_4–CO_2, the already known CH_4–N_2 CIA profiles were suggested to be used instead [4]. Later, Wordsworth et al. [5] conducted theoretical modeling of CH_4–CO_2 CIA assuming that its shape can be approximated as a linear combination of CH_4–CH_4 and CO_2–CO_2 CIA spectra, with coefficients fitted to match the zeroth spectral moment calculated as squared induced dipole averaged over the phase space. This model quantitatively demonstrated that CH_4–CO_2 CIA is more intense than CH_4–N_2 CIA due to the larger polarizability of the CO_2 molecule compared to N_2.

These findings have since stimulated laboratory measurements at room temperature using a Fourier transform spectrometer at the AILES line of the SOLEIL synchrotron facility, beginning with work by Turbet et al. [6] and followed by a second more precise investigation [7]. The measured CH_4–CO_2 CIA coefficients were about 1.6–1.7 times smaller than the theoretical values obtained by Wordsworth et al. [5]. While Godin et al. [8] conducted the first temperature-dependent measurements of CH_4–CO_2 CIA in the 200–300 K range, their results suffer from much larger uncertainty than the room-temperature studies.

To support climate modeling and complement the laboratory measurements, Turbet et al. [7] performed the quantum-mechanical calculations of CH_4–CO_2 CIA within the isotropic approximation. The approach is known [9] to underestimate the true absorption, particularly in case of molecular pairs involving CO_2 exhibiting high anisotropy of interaction. To achieve agreement with experimental measurements, the induced dipole was supplemented with empirical short-range components, fitting the intrinsic parameters of these components. This approach, however, is largely dependent on the quality of the experimental data. The validity of the resulting profiles at temperatures far from room temperature remains therefore uncertain.

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The first fully *ab initio* modeling of CH_4–CO_2 CIA was performed by Fakhardji et al. [10] using classical molecular dynamics simulation, employing the potential energy surface from Hellmann et al. [11] and the long-range induced dipole from Li et al. [12]. While the long-range expansion of the induced dipole is likely to capture the dominant contributions to CIA, neglected short-range components can become significant, particularly at elevated temperatures, as demonstrated by Vigasin [13]. The results of Fakhardji et al. [10] have been included into the HITRAN CIA section in the recent update [14], where they are recommended as the most precise CH_4–CO_2 CIA coefficients available to-date.

In the present work, we advance toward a new level of accuracy in the representation of the CH_4–CO_2 potential energy (PES) and induced dipole (IDS) surfaces, based on *ab initio* calculations under the assumption of rigid monomers. The electronic structure calculations and the approach to PES and IDS fitting are detailed in Section 2. These surfaces were subsequently used in the trajectory-based simulations to compute the CIA spectral profiles. The essential details of our spectral simulation method are outlined in Section 3, with comprehensive description available elsewhere [15,16]. The use of multi-temperature simulation approach enabled calculation of CIA spectra across a broad temperature range from 70 to 400 K with a fine resolution of 10 K step. A significant enhancement, which accelerates the convergence of the Monte Carlo scheme for true dimer component, is introduced in Section 3.1. To rigorously validate our obtained trajectory-based spectra, we made use of independent calculation of the zeroth and second spectral moments performed by means of integration in the phase space with the same PES and IDS, as discussed in Section 3.2. The resulting spectral profiles are presented in Section 4, where we provide a comprehensive comparison with available theoretical and experimental data. An important evidence of accuracy of our obtained spectra came from the excellent agreement with highly accurate measurements in the microwave spectral range (ca 9–15 cm^{-1}) [17]. In contrast, our spectra at room temperature show some divergence with the data from Fakhardji et al. [10] and Turbet et al. [7]. Our method of spectral simulation permits us to obtain CIA spectra at extremely low temperatures, where the contribution from true bound dimers becomes significant. Having these data in hand, we are able to trace the evolution of the dimer absorption profile against the temperature, which is discussed in Section 4.1. Quite expectedly, near room temperature the contribution from true bound dimers to the CIA profile is subtle yet non-negligible, while the absorption by quasibound dimers is clearly manifested, particularly evident in the microwave spectrum [17].

To avoid ambiguity, it is worth clarifying below what is meant under the terms "bound" or "quasibound" dimers. Two molecules held together by weak intermolecular forces are called conventionally "bound", "true bound", or "true" dimers given their total internal kinetic energy is less than the energy required to break an intermolecular bond [18]. In other words, such dimers are definitely trapped inside the potential well which they can leave only if some external energy is supplied by collision with a particle or a photon. Consequently, true dimers can form in a gas only when molecular pair is stabilized by the third collision or emission of photons carrying out an extra energy. The question can be raised, how we manage calculating the spectrum of such true bound dimers through trajectory simulation comprising only pairwise collisions. The answer comes from the special choice of initial conditions restricting the collision dynamics along each selected trajectory by the portion of the phase space which corresponds to bound states only. This can be done because we assume that our system in question is in thermodynamic equilibrium. The rigorous definition of "quasibound" or "metastable" dimers, which are capable to spontaneous decay, is only possible in case of diatomics [19]. For polyatomic molecules we can adopt, e.g., the definition based on the separation of states in approximate energy coordinates [19–21] which was supported by analysis of the scattering trajectory ensemble [22]. Regarding two-particle collisions we can distinguish between trajectories having only one turning point (i.e. an elastic collision) or multiple turning points on each selected trajectory. In this latter case we can claim the formation of a quasibound dimer in between of the first and the last turning points. Detailed discussion of the spectral features of true bound or quasibond dimers which manifest themselves in spectral simulation using trajectory-based approach can be found elsewhere [15,16,23].

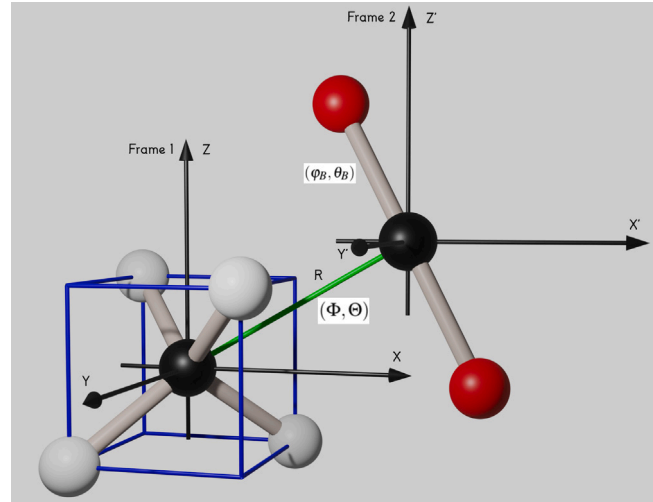

Fig. 1. Schematic representation of the body-fixed coordinate frame for the CH_4–CO_2 complex.

2. Potential energy and induced dipole surfaces

Only a handful of studies have previously investigated the CH_4–CO_2 PES [11,24,25]. Oakley et al. [25] explored interaction energies at the MP2 level of theory. Subsequently, Pai et al. [24] and Hellmann et al. [11] performed more extensive calculations at the CCSD(T) level, using 466 and 707 configurations, respectively. Both studies employed a site-site potential function to represent the PES analytically. Here, we report new calculations of the *ab initio* potential energy and induced dipole surfaces for the CH_4–CO_2 pair.

2.1. Symmetry-adapted basis and grids

Let us consider CH_4 and CO_2 as rigid monomers in their ground electronic and vibrational states. The body-fixed frame, centered on the carbon atom of CH_4, is defined so that methane is static with the vectors from carbon atom to hydrogen atoms having the following coordinates (see Fig. 1),

$$\begin{aligned}
\mathbf{v}_1 &= \frac{1}{\sqrt{3}}\left\{l_{\mathrm{C-H}}, l_{\mathrm{C-H}}, -l_{\mathrm{C-H}}\right\},\\
\mathbf{v}_2 &= \frac{1}{\sqrt{3}}\left\{-l_{\mathrm{C-H}}, -l_{\mathrm{C-H}}, -l_{\mathrm{C-H}}\right\},\\
\mathbf{v}_3 &= \frac{1}{\sqrt{3}}\left\{-l_{\mathrm{C-H}}, l_{\mathrm{C-H}}, l_{\mathrm{C-H}}\right\},\\
\mathbf{v}_4 &= \frac{1}{\sqrt{3}}\left\{l_{\mathrm{C-H}}, -l_{\mathrm{C-H}}, l_{\mathrm{C-H}}\right\},
\end{aligned} \tag{1}$$

where $l_{\mathrm{C-H}}$ is the C–H bond length. Let us represent the vector $\mathbf{R}$ connecting centers of mass of CH_4 and CO_2 using spherical coordinates, $\mathbf{R} = (R, \Phi, \Theta)$. A secondary frame (Frame 2), aligned with the body-fixed frame (Frame 1), is centered on the carbon atom of CO_2. Within this frame, the orientation of the CO_2 axis $\mathbf{l}_{\mathrm{C-O}}$ is given by the spherical coordinates, $\mathbf{l}_{\mathrm{C-O}} = (l_{\mathrm{C-O}}, \varphi_{\mathrm{B}}, \vartheta_{\mathrm{B}})$, where $l_{\mathrm{C-O}}$ is the C–O bond length.

The symmetry properties of the CH_4–CO_2 pair are systematically incorporated into the construction of the angular basis set and the

selection of the grid points, following the approach put forward in our previous effort [26]. Within this approach, the two-body component of potential energy and induced dipole are expanded in terms of symmetry-adapted angular functions. Within the body-fixed frame, the scalar $T^{(0)}_{\{A\}}$ and vectorial $T^{(1)}_{\{A\}m}$ basis functions are given, respectively, by [26]

$$T^{(0)}_{\{A\}}(\Phi,\Theta,\varphi_B,\vartheta_B) = \frac{1}{\sqrt{128\pi^4}}(-1)^{L_A+L_B+L}\sqrt{(2L_A+1)(2L_B+1)(2L+1)}\times \sum_{M_B,M}\begin{pmatrix} L_A & L_B & L \\ M_A & M_B & M\end{pmatrix} C^L_M(\Phi,\Theta)C^{L_B}_{M_B}(\varphi_B,\vartheta_B), \tag{2}$$

and

$$T^{(1)}_{\{A\}m}(\Phi,\Theta,\varphi_B,\vartheta_B) = \frac{1}{\sqrt{128\pi^4}}(-1)^{L_A+L_B+L+\lambda+m}\times \sum_{M_B,M_{AB},M}(-1)^{M_{AB}}\begin{pmatrix} L_A & L_B & L \\ M_A & M_B & -M_{AB}\end{pmatrix}\begin{pmatrix} L & \lambda & 1 \\ M_{AB} & M & -m\end{pmatrix}\times \sqrt{(2L_A+1)(2L_B+1)(2L+1)(2\lambda+1)}C^{\lambda}_M(\Phi,\Theta)C^{L_B}_{M_B}(\varphi_B,\vartheta_B), \tag{3}$$

where $C(\cdot,\cdot)$ is the Racah spherical function and $\begin{pmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot\end{pmatrix}$ is the Wigner 3-j symbol. Also, $\mathbf{L}_A$ is the angular-momentum vector of CH_4 with quantum number L_A and body-fixed projection M_A. Similarly, $\mathbf{L}_B$ for CO_2 has quantum number L_B and body-fixed projection M_B. The angular-momentum quantum number and space-fixed projection of the vector $\mathbf{L} = \mathbf{L}_A + \mathbf{L}_B$ are denoted with L and M_{AB}, respectively. Additionally, the λ and its projection M pertain to the angular momentum λ of the effective diatomic fragment connecting moieties' centers of mass. Finally, the total angular momentum $\mathbf{J} = \mathbf{L} + \lambda$ is characterized by its quantum number J and projection m.

The symmetry-adapted angular basis functions are constructed using the eigenfunction method [27,28], a group-theoretical approach to efficiently perform projection on specific irreducible representations. This method involves diagonalizing a complete set of commuting operators within the target linear space. The resulting eigenvectors constitute the symmetry-adapted basis, with each eigenvector uniquely labeled by the eigenvalues that identify its irreducible representation. For the CH_4–CO_2 pair, the molecular symmetry group is G_{48} [29] — the direct product of methane symmetry group $\mathcal{T}_d(M)$ and the group G_2, which comprises the identity and the operator, that interchanges the oxygen atoms in CO_2. Within the G_{48} group, the potential energy and induced dipole transform under the A_1^+ and A_2^+ irreps, respectively. The complete set of commuting operators was packed into a single linearly composed operator the same way as in Ref. [26], permitting us to project the scalar and vectorial functions with quantum numbers up to $L_A^{max} = 12$, $L_B^{max} = 12$, $L^{max} = 16$ and $\lambda^{max} = 16$ onto the A_1^+ and A_2^+ irreps.

The potential energy and induced dipole were computed at the nodes of Sobolev quadratures, which form two-dimensional grids over the sphere. The quadrature rule for a function f

$$Q_G[f] = \sum_{i=1}^{N_{gen}}\omega_i\sum_{\zeta\in G\zeta_i} f(\zeta) \tag{4}$$

is partitioned into disjoint orbits $G\zeta_i = \{g\zeta_i | g\in G\}$, generated from a representative set $\{\zeta_i, i = 1\ldots N_{gen}\}$, where each point within a given orbit shares an identical weight ω_i. Crucially, both the potential energy and induced dipole remain invariant under the molecular symmetry group operations, meaning their values remain unchanged along any single orbit. This enables the summation in Eq. (4) to be reduced exclusively to the generator set:

$$Q_G[f_{inv}] = \sum_{i=1}^{N_{gen}}\omega_i\|G\zeta_i\| f(\zeta_i), \tag{5}$$

where $\|G\zeta_i\|$ denotes the cardinal number of the orbit $G\zeta_i$. To exploit the molecular symmetry of CH_4–CO_2, we employ the direct product of quadrature generator grids for the tetrahedral (T) and inversion (C_i) groups. The corresponding nodes and weights were adopted from Gräf [30], where they were obtained using the conjugate gradient optimization. Specifically, we opted to use two grids for the T group – comprising 364 and 244 points with corresponding generator sets of $N_{gen} = 31$ and 21 points, respectively – and a grid of 220 points ($N_{gen} = 110$ points) for the C_i group. The larger grid for T group was applied for computing the induced dipole values at intermolecular distances shorter than $R = 8$ a_0, where the interaction is more anisotropic and requires higher angular resolution, while the smaller grid sufficed for larger distances, as well as for sampling the potential energy surface in the whole interval of distances. These quadrature schemes enable exact integration of spherical harmonics with angular momentum up to $l = 16$ and $l = 13$ for the T group, and $l = 12$ for the C_i group.

2.2. Electronic structure calculations

Interaction energies were computed using the explicitly-correlated CCSD(T)-F12 method [31] with the aug-cc-pVTZ basis set, which closely matches CCSD(T)/CBS values at a fraction of the computational cost [32]. Induced dipole values were computed at the CCSD(T)/aug-cc-pVTZ level. We computed interaction energies and induced dipoles for 83 600 configurations and 64 680 configurations, respectively — approximately two orders of magnitude more then previous studies [11, 24].

The interaction energy and induced dipole computations were carried out using the MOLPRO 2010 package [33]. The obtained values were corrected for basis set superposition error (BSSE) using the Boys and Bernardi compensation scheme [34]. The lengths of C–H and C–O were fixed according to the experimental zero-point vibrationally averaged values of 2.067 a_0 and 2.196 a_0, respectively. The size inconsistency due to the use of scaled F12 triple energy correction was corrected for by subtracting the isotropic constant of 13.368 cm^{-1}. The interaction dipole components were computed using the finite-field approach by applying weak homogeneous electric field with strength of 0.001 a.u.

2.3. Least-squares fitting

The determination of expansion coefficients for the angular basis set given by Eqs. (2)–(3) was formulated as a weighted least-squares problem, following the approach detailed in Ref. [26]. The Sobolev quadrature weights were augmented with a Boltzmann-type factor $\xi(U) = \exp(-U/k_B T_{fit})$, where U is the potential energy value and the effective temperature T_{fit} was set to 2000 K. Implementing this weighting scheme allows us to shift the fitting error toward higher-energy configurations, which is particularly advantageous at shorter intermolecular distances where the anisotropy is most pronounced.

The long-range induced dipole surface was constructed to order of R^{-7} using the molecular parameters listed in Table 1, which are calculated at the CCSD(T)/aug-cc-pV5Z level of theory within finite-field approach. The surface is represented as a sum of induction and dispersion contributions [35], with the latter determined in the constant-ratio approximation [36] and employing the isotropic dispersion coefficient $C_6^{000} = 128.7$ taken from Ref. [37]. For reference and comparison, Table 1 also lists the parameters of the induced dipole model from Ref. [12], which was used in the earlier CIA study of Ref. [10]. While the present model retains terms through R^{-7} and therefore includes additional coupling contributions, the model of Ref. [12] (accurate to R^{-6}) provides a more detailed treatment of dispersion contribution through two coefficients: $C_6^{000} = -142.6$ and $C_6^{020} = -38.5$. The long-range surface was sampled on the same Sobolev-type quadrature grid employed for the *ab initio* calculations to enable its decomposition into symmetry-adapted expansion. To ensure consistent asymptotic

Table 1
Molecular properties used to construct long-range induced dipole (in atomic units). The values reported in Ref. [12] are shown, when available, in parentheses.

Property	Definition	CH_4	CO_2
Θ_{zz}	Quadrupole moment, ea_0^2	0	−3.1643 (−3.239)
Ω_{xyz}	Octopole moment, ea_0^3	2.6521 (2.54)	0 (0)
Φ_{zzzz}	Hexadecapole moment, ea_0^4	−7.9547 (−8.27)	−2.4744 (−1.74)
α_{zz}	Polarizability, $e^2a_0^2E_h^{-1}$	16.60 (17.26)	26.6861 (27.140)
α_{xx}		16.60 (17.26)	12.8073 (12.869)
β_{xyz}	First hyperpolarizability, $e^3a_0^3E_h^{-2}$	−9.0184 (−8.09)	0 (0)
$A_{x,yz}$	Dipole–quadrupole polarizability, $e^2a_0^3E_h^{-1}$	9.3368 (9.01)	0 (0)
$E_{z,zzz}$	Dipole–octopole polarizability, $e^2a_0^4E_h^{-1}$	−19.1093 (−18.9)	185.6365 (187.76*)
$E_{x,xxx}$		−19.1093 (−18.9)	−67.7699 (n/a)
$D_{x,yzzz}$	Dipole–hexadecapole polarizability, $e^2a_0^5E_h^{-1}$	−59.9636 (n/a)	0 (n/a)
$D_{x,xxyz}$		112.1830 (n/a)	0 (n/a)
$C_{xx,xx}$	Quadrupole–quadrupole polarizability, $e^2a_0^4E_h^{-1}$	37.2383 (n/a)	33.3387 (n/a)
$C_{zz,zz}$		37.2383 (n/a)	80.6469 (n/a)
$C_{xz,xz}$		32.70748 (n/a)	52.8445 (n/a)
$B_{zz,zz}$	Dipole–dipole–quadrupole polarizability, $e^3a_0^4E_h^{-2}$	−291.7052 (−256)	−294.2696 (−305)
$B_{xz,xz}$		−241.1232 (−219)	−199.2399 (−205)
$B_{xx,zz}$		145.8526 (128)	65.0388 (79)
$B_{xx,xx}$		−291.7052 (−256)	−142.5060 (−140)

*The coefficient is included but does not contribute to the expansion of Ref. [12] through order R^{-6}, as it couples to the quadrupole moment of CH_4 that vanishes identically.

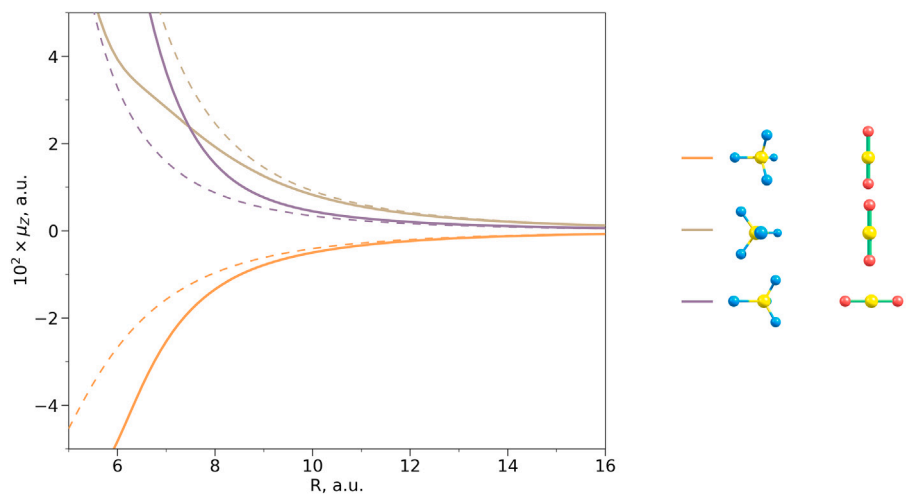

Fig. 2. Radial cross-sections of the CH_4–CO_2 IDS for selected angular orientations. Solid lines refer to *ab initio* dipole, whereas the long-range dipole constructed with parameters listed in Table 1 is denoted with dashed lines.

behavior, the resulting expansion coefficients were incorporated into the fitting routine to correct the values derived from *ab initio* calculated data at intermolecular separations exceeding 18 a_0. As illustrated in Fig. 2 the radial cross-sections of *ab initio* and long-range dipole surfaces converge at large intermolecular distances. The comparison also clearly reveals the magnitude of the short-range components.

As an additional point of reference, Fig. 3 compares radial cross-sections of our potential energy surface with that of Hellmann et al. [11]. While the overall good agreement between the surfaces can be observed, the Hellmann et al. potential yields modestly lower values for some orientations. This can be putatively attributed to the specific adjustment of its parameters to reproduce the experimental temperature dependence of the second virial coefficient. The global minimum of our surface corresponds to a single symmetry-unique equilibrium structure, characterized by a well depth of $U_{\min} = 342.66\ \mathrm{cm}^{-1}$. As summarized in Table 2, this value aligns well with previously reported literature data. To facilitate further research, the functions implementing the smooth representation of our PES and IDS in C and C++, respectively, are provided in the Supplementary Material.

Table 2
Parameters for the minimal energy CH_4–CO_2 configuration.

Method	Well depth (cm^{-1})	Reference
CCSD(T)-F12a	342.66	Present Work
CCSD(T)	347.51	Ref. [11]
CCSD(T)	354.524	Ref. [24]
MP2	202.94	Ref. [25]

2.4. Cross second virial coefficient

The cross second virial coefficient serves as one of the few available benchmarks for assessing the accuracy of the PES. Within the classical approximation, the cross second virial coefficient for CH_4–CO_2 is given by

$$B_{12}^{\mathrm{class}}(T) = -\frac{N_A}{2}\int\left[\exp\left(-\frac{U}{k_\mathrm{B}T}\right)-1\right]R^2\,\mathrm{d}R\,\mathrm{d}\Phi\,\sin\Theta\,\mathrm{d}\Theta\,\mathrm{d}\varphi_\mathrm{B}\,\sin\vartheta_\mathrm{B}\,\mathrm{d}\vartheta_\mathrm{B}, \quad (6)$$

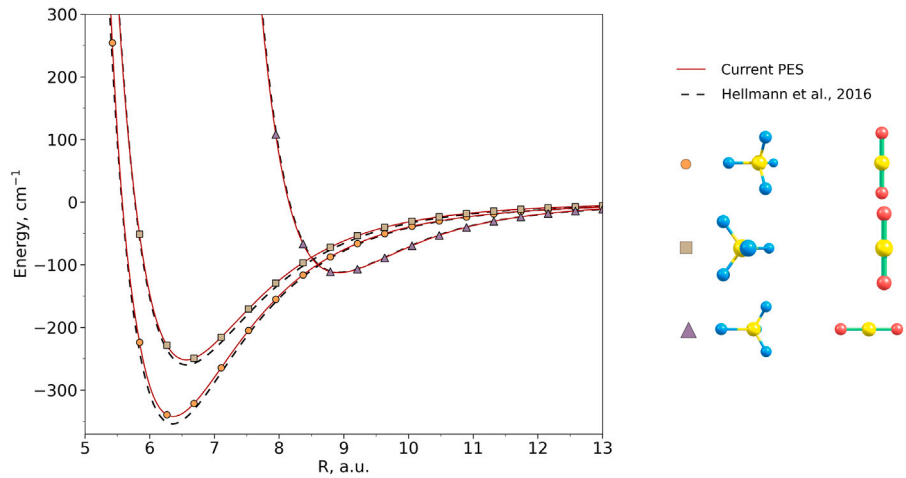

Fig. 3. Radial cross-sections of the CH_4–CO_2 PES for selected angular orientations. The red line shows the values derived from our fit, while the dashed black line corresponds to the surface from Hellmann et al. [11]. Distinct symbols are used to mark curves associated with different orientations of moieties.

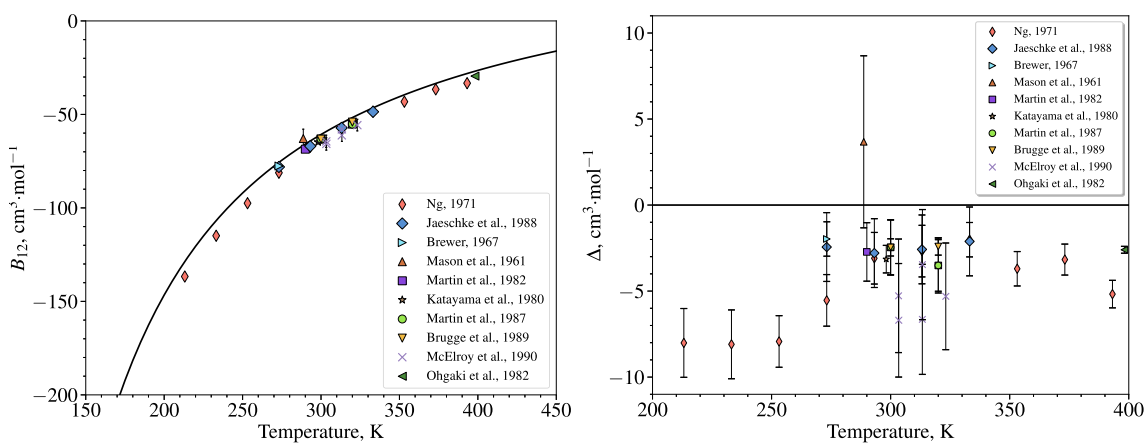

Fig. 4. Temperature variation of the cross second virial coefficient of the CH_4–CO_2 mixture. The left panel compares theoretical values, B_{12}^{class}, calculated from Eq. (6) with the available experimental data, B_{12}^{exp}. The right panels shows the deviation, $\Delta = B_{12}^{\text{exp}} - B_{12}^{\text{class}}$, of experimental values from the calculated values.

where N_A denotes the Avogadro number. We performed integration in Eq. (6) using the adaptive Monte Carlo VEGAS algorithm [38,39]. Fig. 4 demonstrates satisfactory agreement between our calculated values and experimental data taken from reference book of Dymond et al. [40]. We note that the deviation of our calculated results from experimental data is of comparable magnitude to that obtained for the unadjusted potential energy surface by Hellmann et al. (see Figure 3 in Ref. [11]). The complete set of calculated second virial coefficients is provided in the Supplementary Material. We further note that accounting for vibrational degrees of freedom in the CH_4–N_2 system has been shown to improve the agreement with experimental data [41], suggesting a direction for refinement of CH_4–CO_2 interaction model.

3. Trajectory-based approach to modeling CIA spectra

The binary absorption coefficient, $\alpha(\nu)$, for a mixture of gases is defined as [42]

$$\alpha(\nu) = \frac{(2\pi)^4 N_L^2}{3hc}\nu\left[1 - \exp\left(-\frac{hc\nu}{k_B T}\right)\right] G(\nu), \tag{7}$$

where N_L is the Loschmidt number, c is the speed of light, ν is the wavenumber, and h and k_B are Planck and Boltzmann constants, respectively. The spectral function $G(\nu)$ is obtained by Fourier transforming the time-correlation function $C(t)$:

$$G(\nu) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} C(t) e^{-2\pi i c\nu t}\, dt. \tag{8}$$

which is defined as the statistically averaged product of time-shifted dipole moments:

$$C(t) = \frac{V}{4\pi\varepsilon_0}\left\langle \boldsymbol{\mu}(0)\cdot\boldsymbol{\mu}(t)\right\rangle, \tag{9}$$

where V is the gas volume, ε_0 is the vacuum permittivity, and the angular brackets denote phase-space averaging. In the present work, the classical correlation function, denoted as $C_{\text{cl}}(t)$, is estimated using a trajectory-based approach. This function is symmetric in the time domain, and its Fourier transform, consequently, yields a spectral function $G_{\text{cl}}(\nu)$ that is likewise symmetric in frequency. To obtain a spectral profile that adheres to the detailed balance condition [42], we have to employ corrections, which are commonly referred to as desymmetrizations. Several such procedures are available in the literature. Based on our prior analysis of rototranslational CIA bands [16,43], the procedures proposed by Schofield [44]

$$G_{\text{Sch}}(\nu) = G_{\text{cl}}(\nu)\exp\left(\frac{hc\nu}{2k_B T}\right) \tag{10}$$

and Frommhold [42]

$$G_{\mathrm{Fromm}}(\nu) = \frac{C_{\mathrm{cl}}(0)}{C_{\mathrm{cl}}(\beta\hbar/\sqrt{2})} \exp\left(\frac{hc\nu}{2k_{\mathrm{B}}T}\right) \int_{-\infty}^{+\infty} C_{\mathrm{cl}}\left(\sqrt{t^2 + (\beta\hbar)^2/2}\right) e^{-2\pi i c\nu t}\, \mathrm{d}t \tag{11}$$

are anticipated according to our previous experience [15,16,26] to yield the most accurate agreement with experimental data. Though this perspective contrasts with the findings of Fakhardji et al. [10] for CH_4–CO_2 CIA, who identified the procedure due to Bader and Berne [45]

$$G_{\mathrm{Bader}}(\nu) = \frac{\hbar c\nu}{k_{\mathrm{B}}T} \frac{1}{1 - \exp\left(-hc\nu/k_{\mathrm{B}}T\right)} G_{\mathrm{cl}}(\nu) \tag{12}$$

as providing superior agreement with the experimental data from Turbet et al. [7]. It is worth reminding that modeling of Titan's radiance spectra achieved the best agreement with Cassini spacecraft observations through a semi-empirical model combining 35% Schofield's and 65% Frommhold's corrections [43]. We can speculate that the CH_4–CO_2 case can likely be similar to CH_4–N_2 in this concern. However, it should be noted that Schofield's procedure tends to overestimate absorption and can introduce unphysical artifacts in the far wing at lower temperatures [10,46].

The following section introduces a computational approach that accelerates the trajectory-based simulation of contribution due to bound states.

3.1. Accelerating convergence for bound states' contribution

In our trajectory-based approach, the dipole autocorrelation function is directly estimated as [16]:

$$C_N(t_j) = \frac{V}{4\pi\varepsilon_0} \frac{1}{N} \sum_{i=1}^{N} \boldsymbol{\mu}_i(0) \cdot \boldsymbol{\mu}_i(t_j) \exp\left(-\frac{E_i}{k_{\mathrm{B}}T}\right), \tag{13}$$

where the index i runs over calculated trajectories, $\boldsymbol{\mu}_i(t)$ is the induced dipole at time t in the ith trajectory, Δt is the sampling interval, $t_j = j\Delta t$, and E_i is the energy for the ith trajectory. In molecular dynamics simulations, the expression is typically augmented by averaging over time origins (see Refs. [47,48]):

$$\boldsymbol{\mu}_i(0) \cdot \boldsymbol{\mu}_i(j\Delta t) \longrightarrow \frac{1}{M+1} \sum_{k=0}^{M} \boldsymbol{\mu}_i(k\Delta t) \cdot \boldsymbol{\mu}_i(j\Delta t + k\Delta t), \tag{14}$$

where $M + 1$ is the number of sampled time points. This time-origin averaging is valid in molecular dynamics simulation because periodic boundary conditions maintain a time-invariant canonical distribution for the prepared ensemble of molecules. However, this approach is not universally applicable to trajectory-based simulation. Here, when trajectories corresponding to unbound domain are considered, initial conditions at $t = 0$ confine molecular pair within a sphere of radius $R_{\max}$ (see the algorithm of initial conditions preparation in Sec IV.B of Ref. [16]). As the trajectory is propagated, pair is no longer confined to initially chosen sphere, thus violating the spacial periodicity required for time-origin averaging. In contrast, molecules in a pair that belongs to bound domain remain in proximity to each other, which, in turn, permits an efficient reformulation of the correlation function as

$$\tilde{C}_N(t_j) = \frac{V}{4\pi\varepsilon_0} \frac{1}{N} \sum_{i=1}^{N} \frac{1}{M+1} \sum_{k=0}^{M} \boldsymbol{\mu}_i(k\Delta t) \cdot \boldsymbol{\mu}_i(j\Delta t + k\Delta t) \exp\left(-\frac{E_i}{k_B T}\right). \tag{15}$$

The calculation for a correlation function $\tilde{C}_N(t)$ in a range $[0, t_{\max}]$ proceeds as follows:

1. Propagate each of N trajectories to a time of $2t_{\max}$;
2. For each time $t \le t_{\max}$, compute the single-trajectory contribution to $\tilde{C}(t)$ using the right-hand side of Eq. (14);
3. Obtain the final estimate for $\tilde{C}_N(t)$ as Boltzmann-weighted average over N single-trajectory contributions.

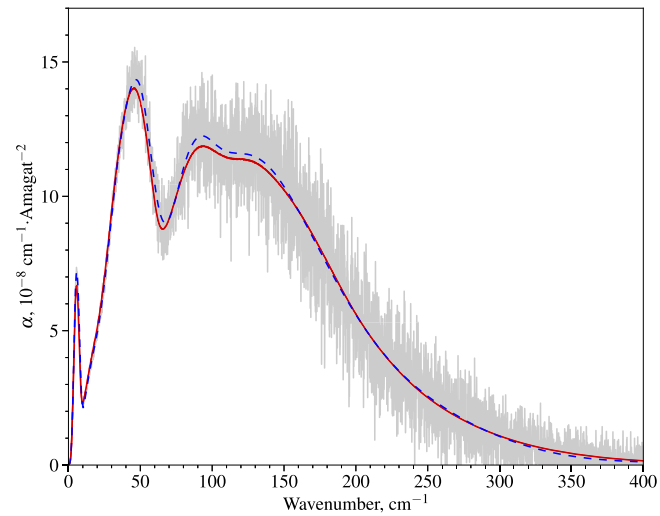

Fig. 5. Bound states' contribution to the CH_4–CO_2 CIA spectrum at 300 K. The red line shows the spectrum obtained from accelerated estimate $\tilde{C}(t)$. The light gray line shows the raw spectrum profile originating from the direct estimate for $C(t)$, whereas the blue dashed line corresponds to smoothed spectrum (see text for details).

We emphasize that each $\tilde{C}(t_j)$ is estimated from $M+1$ time origins since $(j+k)\Delta t$ never exceeds $2t_{\max}$, yielding a statistically robust and balanced estimate at each time point.

Fig. 5 illustrates the application of the convergence-accelerating procedure to computing the CH_4–CO_2 bound states' CIA spectrum at 300 K. First, we conducted trajectory-based simulation to directly estimate bound states' correlation function $C_N(t)$ via Eq. (13), utilizing an ensemble of approximately 700,000 trajectories. This correlation function was subsequently Fourier-transformed into the corresponding spectral function, following the methodology detailed in Ref. [16]. To mitigate numerical noise, the resulting spectrum was smoothed using the LOESS algorithm [49], which employs a weighted local regression over an adaptively sized window. Second, the accelerated estimate $\tilde{C}_N(t)$ was calculated from a separate, twice as small ensemble of approximately 350,000 trajectories.

As illustrated in Fig. 5, the spectrum derived from the accelerated estimate is inherently smooth and exhibits a marked reduction in numerical noise. It is noteworthy that the smoothed spectrum from the direct method aligns closely with the unsmoothed spectrum obtained from the accelerated estimate. This compelling agreement indicates that the convergence-accelerating procedure itself effectively suppresses numerical noise. Consequently, for moderate-sized trajectory ensembles (on the order of one million trajectories), the convergence acceleration method largely obviates the need for post-processing smoothing for bound states' spectra.

3.2. Convergence validation via spectral moment analysis

The CIA band can be quantitatively characterized using its lower-order spectral moments, which describe the intensity distribution within the band [42]. These moments are particularly valuable for validating the convergence of the Monte Carlo algorithm in spectral calculations using trajectory-based approach, as they can be derived from two independent and complementary approaches. The first approach calculates the spectral moments directly from the simulated classical binary absorption coefficient, $\alpha_{\mathrm{cl}}(\nu)$, according to the following expression [42]

$$\mathcal{M}_n^{\mathrm{cl}} = \begin{cases} 2\displaystyle\int_0^{+\infty} \nu^{n-1}\left[1-\exp\left(-\frac{hc\nu}{k_{\mathrm{B}}T}\right)\right]^{-1}\alpha_{\mathrm{cl}}(\nu)\,\mathrm{d}\nu, \\ 0, \end{cases} \tag{16}$$

where the first and second line correspond to even and odd values of n, respectively. Alternatively, the zeroth and second spectral moments can be obtained through phase-space averaging, independent of trajectory ensemble in spectral simulation, as [42]

$$\begin{aligned} M_0 &= \frac{(2\pi)^4 N_L^2}{3h}\frac{V}{4\pi\varepsilon_0}\frac{1}{2\pi c}\langle\boldsymbol{\mu}(0)\rangle^2, \\ M_2 &= \frac{(2\pi)^4 N_L^2}{3h}\frac{V}{4\pi\varepsilon_0}\frac{1}{(2\pi c)^3}\langle\dot{\boldsymbol{\mu}}(0)\rangle^2. \end{aligned} \tag{17}$$

Useful relationships concerning classical and quantum calculations of the first spectral moments can be found in Refs. [16,46]. Fig. 6 presents a comparison of the zeroth and second spectral moments calculated via phase-space averaging (Eq. (17)) and via frequency-domain integration using Eq. (16). Because the classically derived absorption coefficient and the theoretical spectra from Refs. [7,10] adhere to different detailed balance conditions, a direct comparison of their spectral moments requires care (a detailed discussion is provided in Section 4.C of Ref. [16]). To facilitate a consistent comparison, we applied corresponding *symmetrization* procedures – the inverse of their desymmetrization counterparts – using the procedure of Bader and Berne [45] for the data of Fakhardji et al. [10] and that of Schofield [44] for the data of Turbet et al. [7]. The spectral moments for profiles from Ref. [5] were calculated directly without adjustment, as the spectral model was fitted to the classically derived spectral moments and does not conform to the quantum detailed balance condition.

Across the 200–400 K range, our phase-space values for the zeroth spectral moment show close agreement with those derived from the spectral profiles of Refs. [5,7,10]. However, a marked divergence emerges at lower temperatures, where our calculated spectral moment exhibits a more rapid increase. At 100 K, this discrepancy reaches 33% relative to Fakhardji et al. [10] and 60% relative to Turbet et al. [7]. It should be noted that the magnitude of this deviation is comparable to the contribution from bound states to the spectral moment, underscoring its significance in the low-temperature regime. Furthermore, our calculated second spectral moment is consistently larger than those derived from both Fakhardji et al. [10] and Turbet et al. [7] across the entire temperature range, indicating that our calculated CIA coefficients systematically exceed these theoretical models. The second spectral moment derived from Wordsworth et al. [5] exceeds our values approximately 2.6–3 times between 200 and 350 K. This substantial discrepancy arises because their semiempirical model was fitted to reproduce the zeroth moment while the second spectral moment remained unconstrained.

To ensure reliable estimates of the correlation functions, we employed a staged multi-temperature calculation incorporating up to ten reference temperatures (see Ref. [16] for details of adopted multi-temperature reweighting approach). This approach yielded consistency between the trajectory-based spectra and direct phase-space integration, with differences for the zeroth and second spectral moments not exceeding 1.5% and 1.2%, respectively. Reaching this level of convergence required accumulating on the order of 15 to 40 million trajectories for unbound states across the temperature range, while bound states' contribution was captured with 200,000 to 400,000 trajectories. The simulations utilized a sampling rate of 4.29 fs, with trajectories propagated to 281 ps. Additional computational parameters followed the implementation described in Ref. [43]. All trajectory-based CIA calculations, subsequent spectral processing, and phase-space integration were performed using the `Hawaii Hybrid` program, a computational suite which will be presented elsewhere [50].

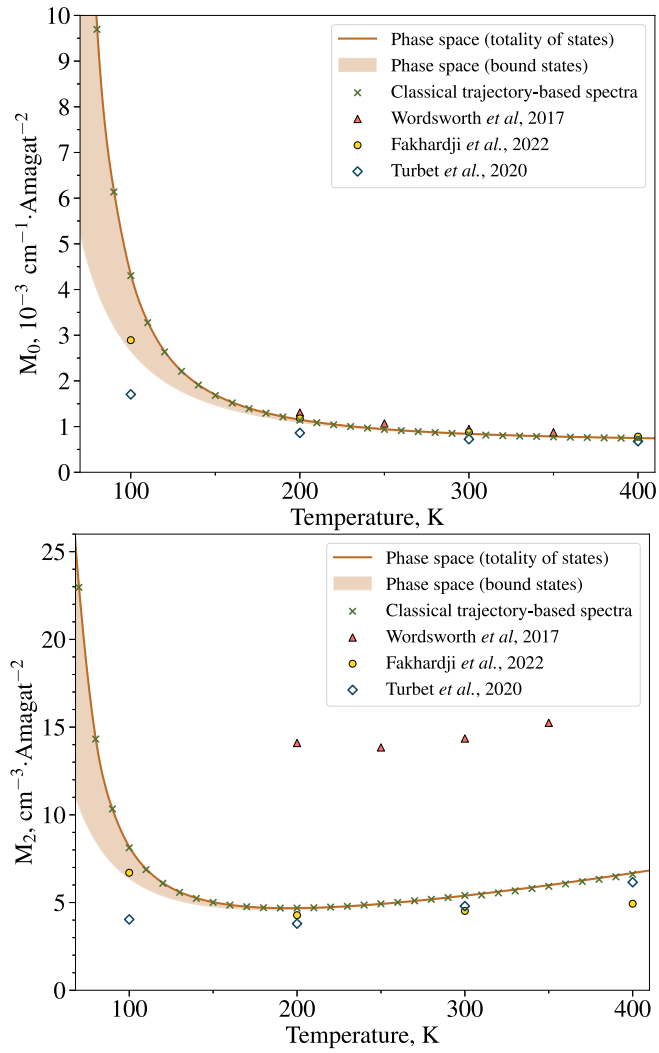

Fig. 6. Calculated temperature variations of zeroth (upper panel) and second (lower panel) spectral moments of the CH_4–CO_2 rototranslational band. The solid line refers to the values obtained through phase-space integration for totality of states. The shaded area indicates the bound states' contribution. The crosses show the values derived from the trajectory-based profiles.

4. Results

We have performed extensive trajectory-based calculations of CH_4–CO_2 CIA using a staged multi-temperature approach across the 70–400 K range with 10 K resolution. The complete set of spectral profiles, formatted according to HITRAN CIA standard (see Karman et al. [51] and HITRAN CIA website[1] for format specification), is provided in the Supplementary material. To quantify the uncertainty associated with desymmetrization, we include spectral profiles obtained with both Schofield's and Frommhold's desymmetrization procedures.

Our calculated trajectory-based spectra at 300 K are compared with available experimental data in Figs. 7 and 8. As shown in Fig. 7, the calculated spectrum exhibits a consistent two-peak structure – including a pronounced secondary feature ca. 300 cm^{-1} – irrespective of whether Schofield's or Frommhold's desymmetrization procedure is employed. The position of this secondary peak aligns reasonably

[1] https://hitran.org/cia/

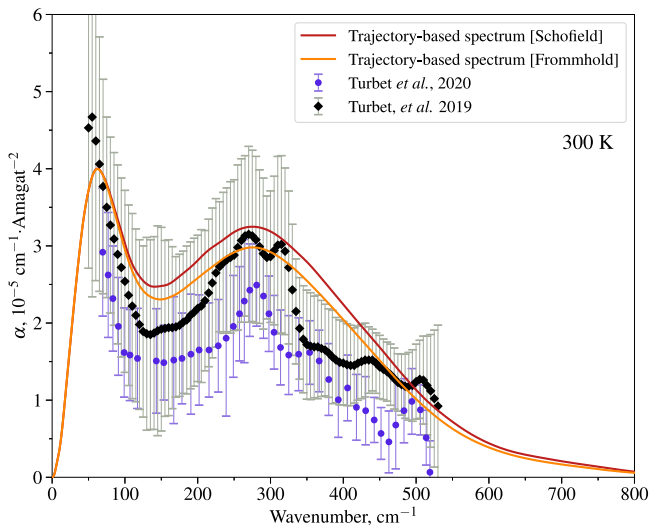

Fig. 7. CH_4–CO_2 collision-induced absorption spectrum at 300 K. The red and orange lines show the trajectory-based spectrum obtained using Schofield's and Frommhold's desymmetrization procedures, respectively. Black and blue dots with error bars denote experimental measurements by Turbet et al. [6] and Turbet et al. [7], respectively.

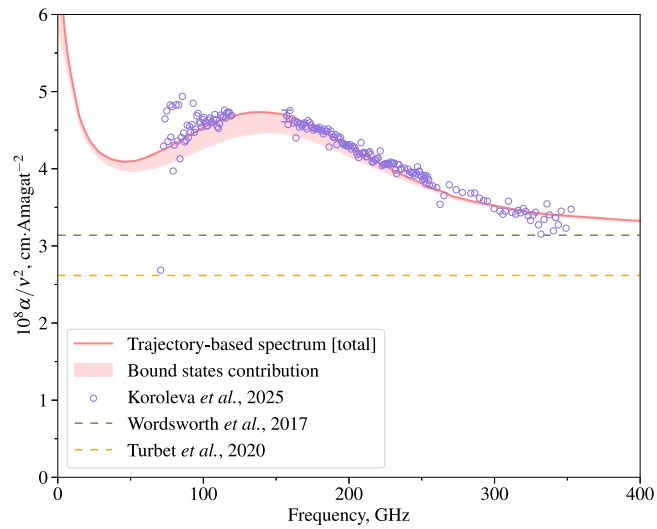

Fig. 8. Reduced spectral function for CH_4–CO_2. The solid line represents the spectral function corresponding to the totality of states, whereas the shaded area shows the contribution from bound states. Blue circles show recent microwave data from Ref. [17]. The dashed green and yellow lines correspond to the theoretical spectra from Wordsworth et al. [5] and Turbet et al. [7], respectively (see text for details).

with recent measurements [6,7]. However, in terms of absolute intensities, our results show closer agreement with the earlier data from Ref. [6], falling outside the error margins reported in Ref. [7]. The choice of desymmetrization procedure introduces an uncertainty of approximately 8% in the intensity of the secondary peak.

Further comparison with other theoretical studies [5,6,10] is presented in Fig. 9. Panel (c) reveals significant discrepancies among the theoretical results at room temperature. Notably, the spectrum from Fakhardji et al. [10] does not exhibit the secondary peak present in experimental results. This absence can be attributed to the lack of short-range components in the induced dipole model since the long-range induced dipole from Ref. [12] was employed within the classical molecular dynamics simulation performed in Ref. [10]. This interpretation is corroborated by the results of Turbet et al. [7], who included empirical short-range components into induced dipole model and successfully reproduced the secondary peak to obtain agreement with the experimental measurements. The theoretical model of Wordsworth et al. [5] significantly overestimates results from other theoretical studies in the vicinity of the secondary peak.

Fig. 8 demonstrates excellent agreement between the calculated reduced spectral function, $\tilde{J}(\nu) = \alpha(\nu)/\nu^2$, and precise measurements conducted using a resonator spectrometer [52] at room temperature with the total pressure reaching 2 atmospheres, following the same approach as in our previous works on CO_2 [53], N_2 [54,55], and N_2–Ar bimolecular absorption [54]. To retrieve CIA component, resonant CH_4 rotational lines, arising from the centrifugal distortion moment [56], were subtracted from the measured absorption using the Van Vleck-Huber profile [57] with HITRAN2020 parameters [58]. The broadening coefficients of CH_4 lines by CO_2 were obtained by scaling the HITRAN air-broadening coefficients by a factor of 1.3, following Tran et al. [59]. The contribution of CH_4 lines to the observed absorption is no more than 0.15%, which enables us to consider questions of line parameters accuracy and line wing shapes as secondary for this analysis. Bimolecular absorption from pure CO_2 and CH_4 was accounted for using the results from Galanina et al. [53] and Koroleva et al. [17], correspondingly, with continuum uncertainty from spectral parameters remaining within 1.5%. The expected pressure dependence of CH_4–CO_2 CIA was validated by the recordings at varying CH_4 and CO_2 mixing ratios, with Fig. 8 representing the average across all experimental datasets. Additional details regarding CIA retrieval and uncertainty analysis can be found in Ref. [17]. A discernible bound states' contribution, manifesting as the dimer R-branch, is evident in the microwave range despite its modest integrated intensity. This level of spectral detail is absent in the data of Turbet et al. [7] and Wordsworth et al. [5], whose coarse spectral resolution of 5 cm^{-1} yields only a few data points in the frequency range shown. Consequently, a parabolic fit to their data shortward of 20 cm^{-1} therefore results in a constant value of $\tilde{J}(\nu)$, entirely lacking the spectral feature that is both observed experimentally and faithfully reproduced by our calculations. The clear signature of quasibound states is particularly evident bearing in mind that contribution due to free states is virtually flat [53]. The impressive agreement between our trajectory-based results and the microwave-range experimental data for CH_4–CO_2 is consistent with our prior analyses of CO_2–Ar and CO_2–CO_2 systems [53,60]. This consistency reinforces the thesis that nature of continuum is established in the microwave range, and moreover inspiring confidence in our results for the entire absorption band. Furthermore, the presence of the Q-branch in our calculated reduced spectral function is anticipated, as the spherical symmetry of CH_4 renders it effectively an atom, making the CH_4–CO_2 pair a direct analogue to CO_2–Ar [16].

Between 200 K and 400 K, the CH_4–CO_2 CIA spectrum exhibits two primary features, as shown in Fig. 9. The first is a peak resulting from absorption in CO_2 molecule induced by the electric multipole field of methane. The second constitutes a broad feature arising from the CO_2 molecule polarizing the methane molecule. The relatively weak anisotropy of CH_4 means its induction effect on CO_2 is accurately captured even by an isotropic model, which explains the strong agreement between our trajectory-based results with the Turbet et al. [7] and Fakhardji et al. [10] spectra in the region of the CO_2-induced peak. In contrast, a substantial discrepancy emerges in the spectral region associated with CH_4 induction by CO_2, due to the highly anisotropic nature of the CO_2 influence. This anisotropy is faithfully represented in our trajectory-based simulations due to the complexity of used induced dipole model, whereas it is not accounted for in the calculations by Turbet et al. [7] and Fakhardji et al. [10]. Consistent with the spectral moment analysis presented in Section 3.2, Wordsworth et al. [5] spectra

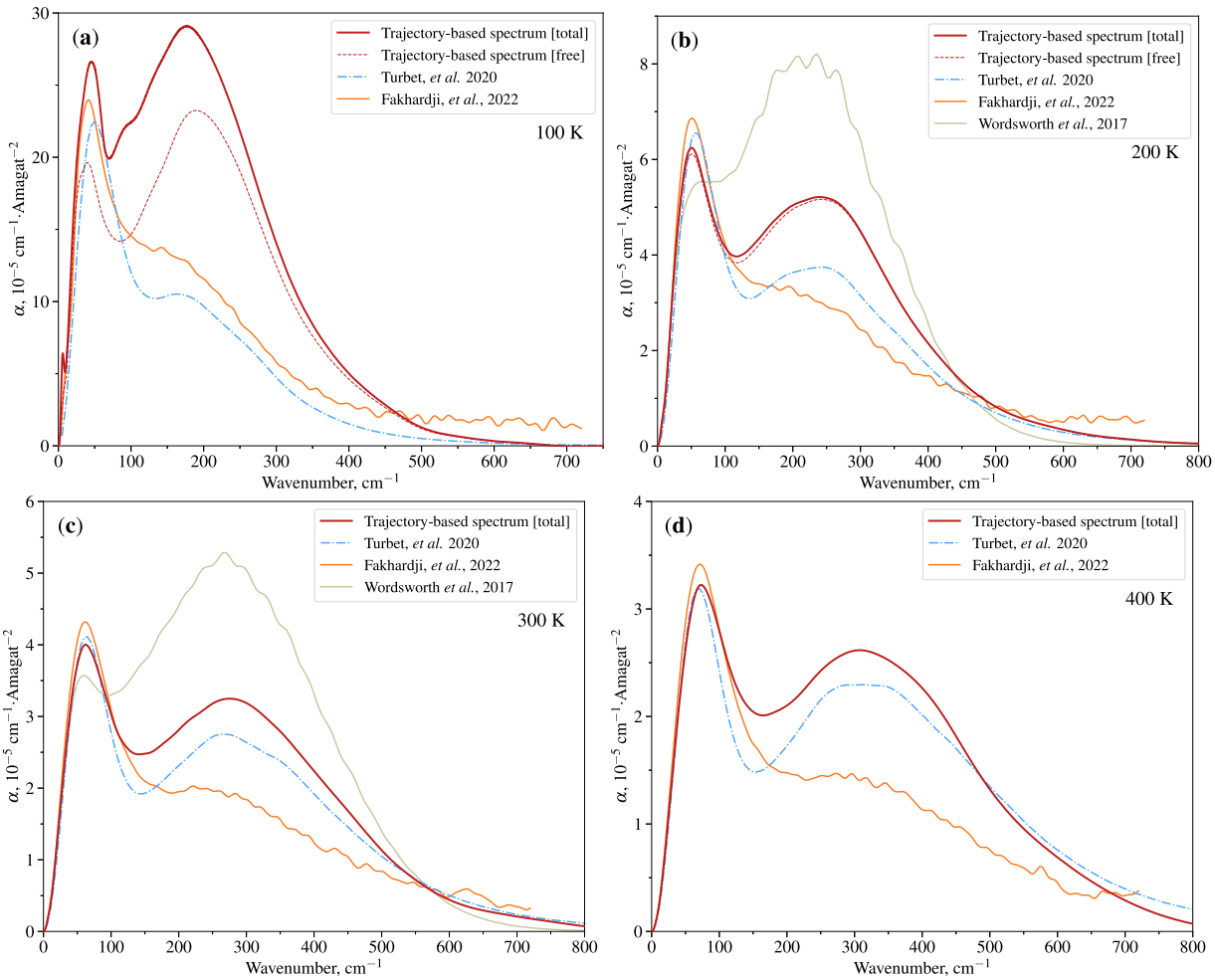

Fig. 9. CH_4–CO_2 collision-induced absorption spectra at selected temperatures. The trajectory-based spectral profiles, obtained using Schofield's desymmetrization procedure, are shown in red, with solid and dashed lines corresponding to the totality of states and free states, respectively. The solid orange line and blue dashed line correspond to the theoretical results from Fakhardji et al. [10] and Turbet et al. [7]. The gray line shows the model spectrum from Wordsworth et al. [5].

substantially exceed our trajectory-based spectra in intensity within the critical range near the secondary peak.

4.1. Manifestation of true dimer contribution in the low-temperature spectra

The discrepancy between our simulated spectral profiles and those from Fakhardji et al. [10] is minimal within the 200–300 K temperature range but increases significantly while approaching both higher and lower ends of the temperature interval. At elevated temperatures, the growing disagreement stems likely from the failure to include the short-range components of induced dipole adopted in Ref. [10], leading to an underestimation of absorption. At lower temperatures, the observed differences can be primarily attributed to insufficient sampling of bound states – a recognized limitation of molecular dynamics simulations of weakly interacting systems – enhanced by the aforementioned deficiencies in the induced dipole model at close intermolecular distances. One cannot fail to mention an additional source of uncertainty coming from the choice of desymmetrization procedure, which also influences the quantitative comparison between the models. The spectral profiles obtained using both Schofield's and Frommhold's procedures are shown in Fig. 10, revealing a variation in the secondary peak intensity of approximately 25%. Upon closer examination, it becomes evident that at temperatures below approximately 100 K, the contribution from bound states gradually increases, giving rise to distinct additional spectral features.

To elucidate the complex spectral features arising from the true dimer absorption, we performed trajectory-based simulation of the CH_4-CO_2 dimer spectra at 5 and 10 K (see Fig. 11). The true dimer absorption band near 100 cm^{-1}, which is already perceptible at 100 K (see Fig. 10), becomes markedly more pronounced in these low-temperature spectra. Furthermore, the broad feature near 50 cm^{-1} resolves into three distinct dimer bands located at roughly 46, 61, and 68 cm^{-1}. Weak overtone and combination bands of intermolecular vibrations can also be distinguished within the 125–200 cm^{-1} region of the trajectory-based spectrum.

For a preliminary assignment of the peaks observed in the dimer spectrum, we calculated the CH_4–CO_2 dimer vibrational frequencies at the equilibrium configuration. The intermolecular frequencies were computed within the harmonic approximation using Wilson's GF-method [61]. The G-matrix was derived from the spaced-fixed classical kinetic energy employed in our trajectory-based simulation. The relevant expression for the kinetic energy can be found in Eq. 12 of Ref. [43]. The F-matrix was obtained by numerical differentiation of our fitted potential energy surface at its global minimum. The positions of the five harmonic intermolecular frequencies are indicated by dashed vertical lines in Fig. 11. As expected, the band origins of the dimer spectrum gradually converge toward the corresponding harmonic frequencies as the temperature approaches zero.

A certain caution is warranted in the quantitative interpretation of the spectra obtained at 5 and 10 K. At such extremely low temperatures,

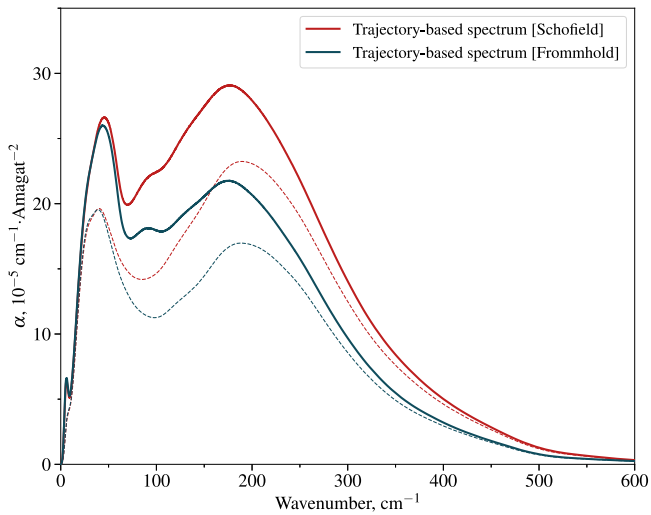

Fig. 10. Calculated CH_4–CO_2 collision-induced absorption spectrum at 100 K. The red and blue lines show the trajectory-based spectra obtained using the desymmetrization procedures of Schofield and Frommhold, respectively. For each, the solid and dashed lines represent the spectra derived from the totality of states and free states only.

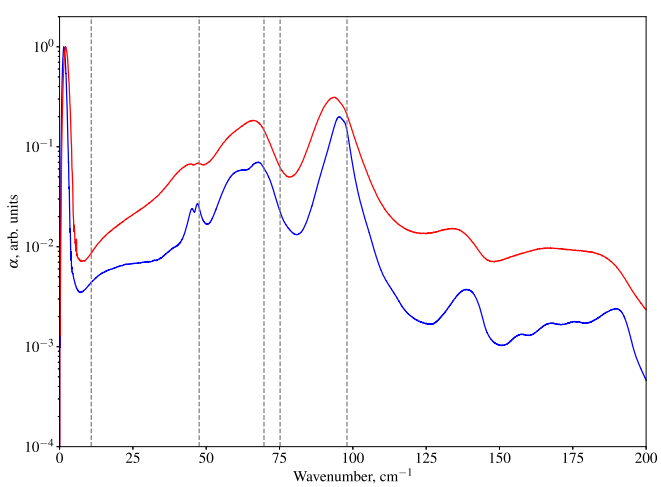

Fig. 11. Calculated CH_4-CO_2 collision-induced absorption spectra for 5 K (blue line) and 10 K (red line), normalized by their respective intensity maxima. The dashed vertical lines indicate the five harmonic vibrational frequencies calculated for the CH_4-CO_2 dimer at its equilibrium geometry.

the classical trajectory-based method encounters limitations, due to the increasingly prominent quantum nature of the nuclear motion. The *ad hoc* desymmetrization procedures typically employed to impose the quantum detailed balance can yield unphysical intensities in this regime. Consequently, the spectra shown in Fig. 11 are purely classical and presented without desymmetrization correction. While these spectra are reported here as an illustrative material, they remain meaningful for identifying approximate band positions. Further analysis of the true dimer spectra obtained from trajectory-based simulation will be reported elsewhere.

5. Conclusions

We have presented a comprehensive investigation of the CH_4–CO_2 collision-induced spectra across a climatically relevant temperature range of 70 to 400 K. The salient feature of our approach lies in its foundation upon high-fidelity potential energy and induced dipole surfaces, constructed from an extensive database of quantum chemical calculations. A notable achievement is the successful and detailed modeling of the bound states' contribution, which becomes increasingly significant at lower temperatures. Furthermore, we introduced a computational technique that substantially accelerates the convergence of contribution due to bound states in our trajectory-based simulation.

The reliability of the resulting spectra is firmly established through rigorous validation against independent theoretical results [7,10], as well as experimental measurements [6,7]. The excellent agreement with highly precise microwave measurements [17], obtained using resonator spectrometer, particularly corroborates the results of our simulations. The close correspondence between the spectral moments derived from the trajectory-based spectral profiles and their phase-space counterparts highlights strong internal consistency of our approach. In contrast to the results from Fakhardji et al. [10], which are recommended in the recent HITRAN CIA update [14], our spectra exhibit the secondary peak in the 200–600 cm^{-1} spectral range that forms the transparency window of CO_2-dominated atmosphere. Although both sets of experimental data from Refs. [6,7] confirm the presence of this feature, our simulated spectral profiles show closer agreement with the earlier measurements from Turbet et al. [6]. Across the studied temperature range, our trajectory-based spectra systematically yield higher intensities compared to the theoretical data from Turbet et al. [7] and Fakhardji et al. [10]. Consequently, the potential contribution of CH_4–CO_2 CIA to the greenhouse warming of the Martian paleoatmosphere requires reevaluation. Given the significant discrepancies among theoretical results and the considerable uncertainties in existing experimental measurements, the need for new precise laboratory studies of CH_4–CO_2 CIA to resolve these issues is clearly recognized.

CRediT authorship contribution statement

A.A. Finenko: Writing – original draft, Investigation, Data curation. **D.N. Chistikov:** Writing – original draft, Software, Investigation. **A.O. Koroleva:** Investigation, Formal analysis, Data curation. **A.A. Vigasin:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

The supplementary material provides the calculated CH_4–CO_2 binary absorption coefficients from 70 to 400 K in 10 K increments, tabulated in the HITRAN CIA format. Also included are the complete set of calculated second virial coefficients covering the same temperature range. Additionally, we provide routines in C and C++ that implement smooth approximations for the CH_4–CO_2 PES and IDS accompanied by the example drivers.

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jqsrt.2026.109810.

Data availability

Data will be made available on request.

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